Predicting with distributions

Predicting with distributions

Suppose we have two variables with normal distribution,

- Home team score
- Away team Score

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$$A \sim N\left(\mu_A, \sigma_A^2\right)$$

Then the difference between the home and away teams scores will also have normal distribution with the following parameters (assuming this two are independent)

$$\mu_D = \mu_H - \mu_A$$
$$\sigma_D^2 = \sigma_H^2 + \sigma_A^2$$

Thus the difference will have a normal distribution with the following parameters

$$D \sim N\left(\mu_H - \mu_A, \sigma_H^2 + \sigma_A^2\right)$$

Predicting with distributions

- In NBA (National Basketball Association) points scored per game by home and away teams follow normal distribution.
- Install package SportsAnalytics270 from github.
- The package contains data for Sports analytics

```
library(devtools)
install_github("HABET/CSE270")
library(SportsAnalytics270)
```

```
library(SportsAnalytics270)
```

Predicting with distributions

- dataframe nba2009_2018 contains NBA regular season games from 2009 to 2018-2019 season

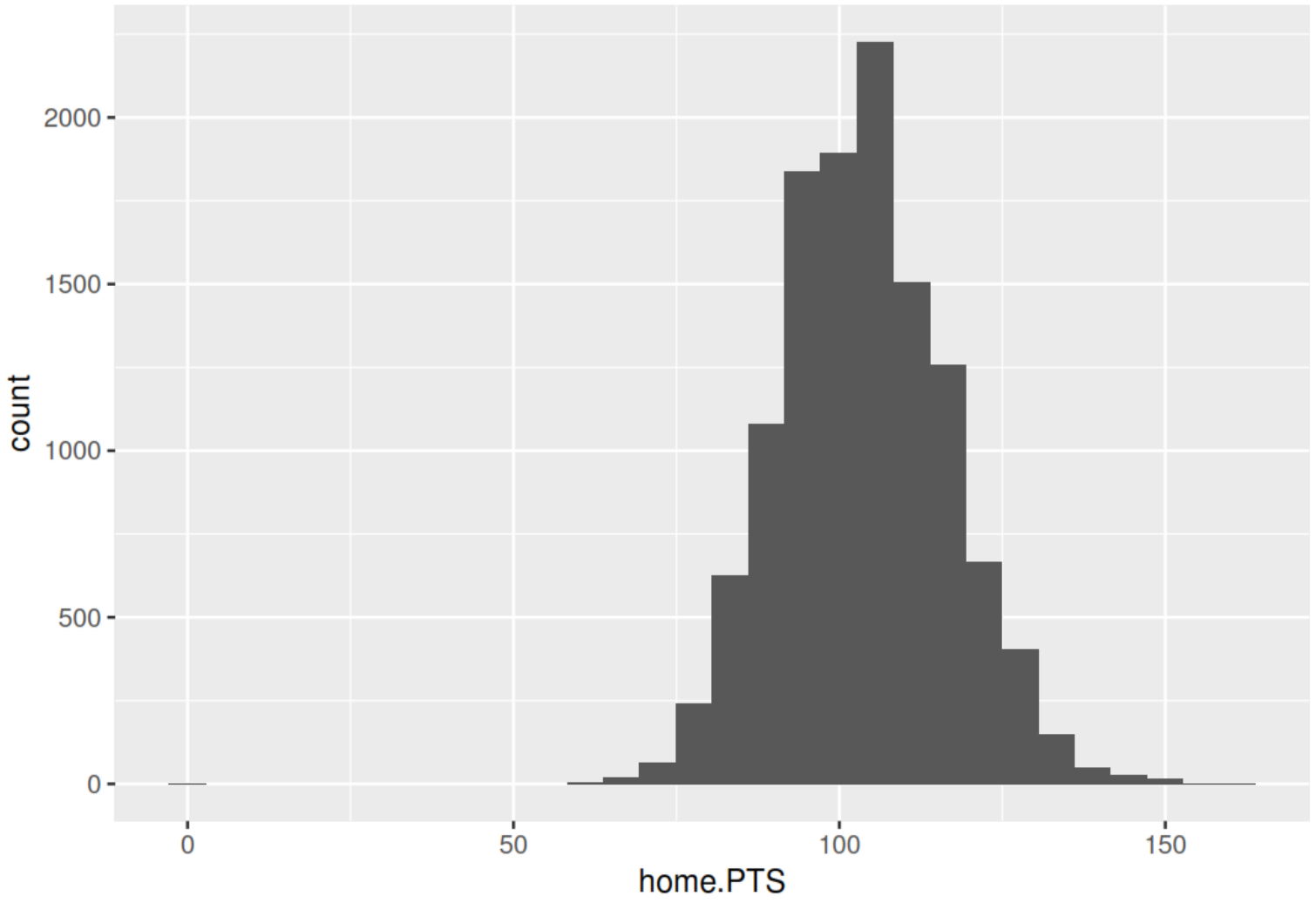
```
data(nba2009_2018)
head(nba2009_2018)
```

```
##   SEASON_ID  GAME_DATE home.TEAM_ABBREVIATION         home.TEAM_NAME
## 1      2009 2009-10-27                     CLE    Cleveland Cavaliers
## 2      2009 2009-10-27                     DAL       Dallas Mavericks
## 3      2009 2009-10-27                     POR Portland Trail Blazers
## 4      2009 2009-10-27                     LAL     Los Angeles Lakers
## 5      2009 2009-10-28                     ATL          Atlanta Hawks
## 6      2009 2009-10-28                     ORL          Orlando Magic
##   home.PTS away.TEAM_ABBREVIATION     away.TEAM_NAME away.PTS home.WL
## 1       89                     BOS     Boston Celtics       95       L
## 2       91                     WAS Washington Wizards      102       L
## 3       96                     HOU    Houston Rockets       87       W
## 4       99                     LAC        LA Clippers       92       W
## 5      120                     IND     Indiana Pacers      109       W
## 6      120                     PHI Philadelphia 76ers      106       W
```

Predicting with distributions

Plot the histogram for home points

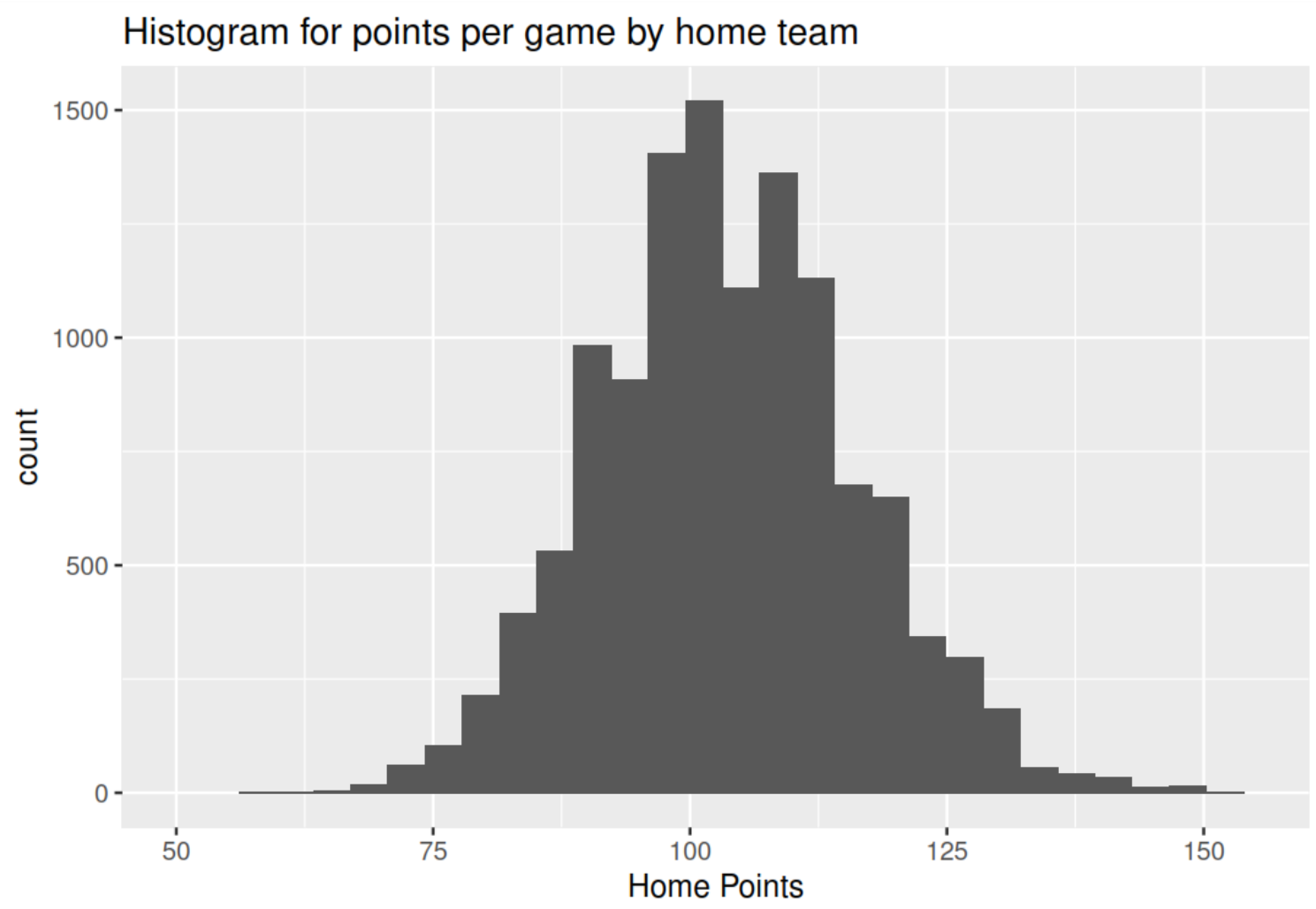
```
library(ggplot2)
ggplot(nba2009_2018, aes(x=home.PTS))+geom_histogram()
```



Predicting with distributions

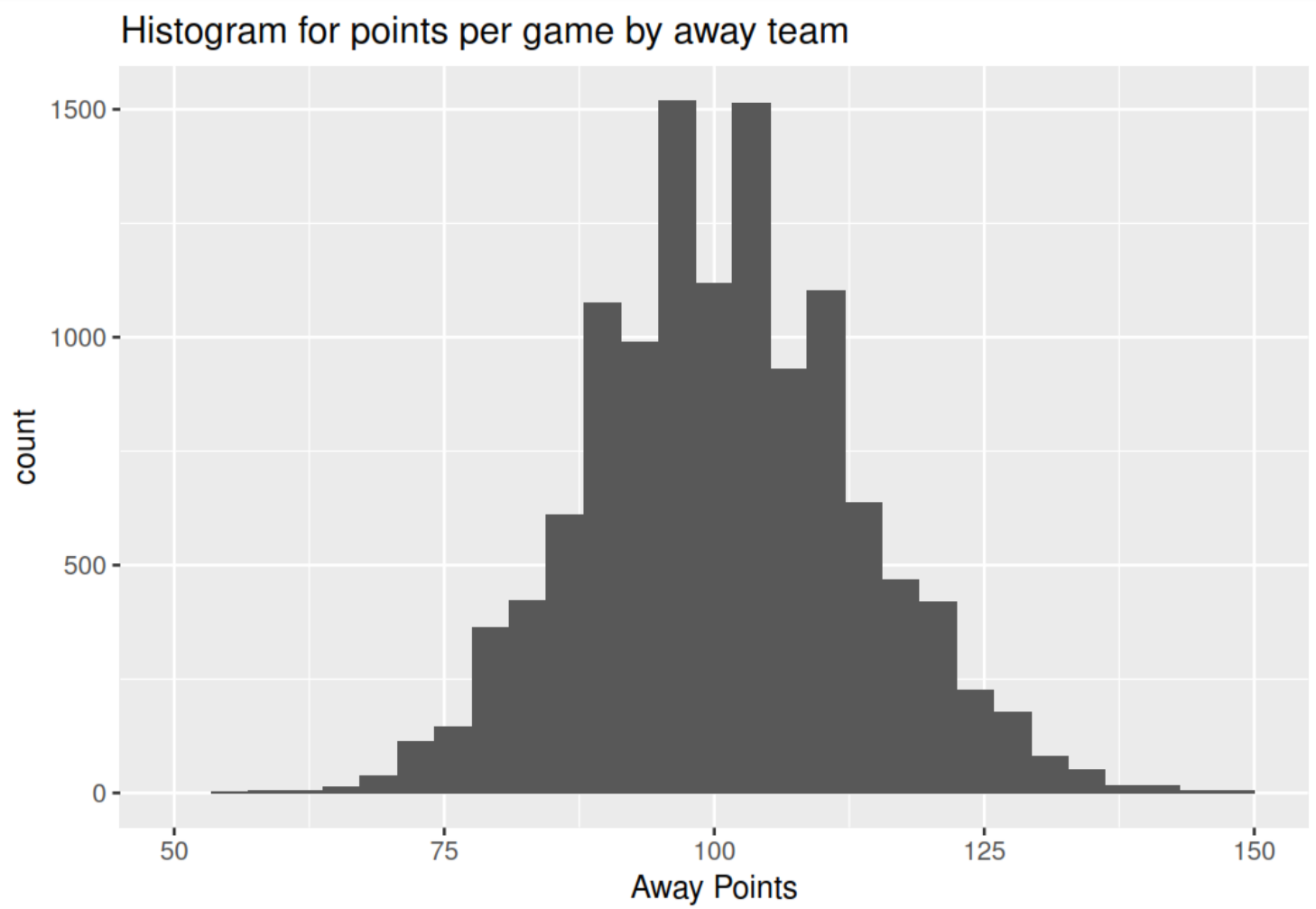
Exclude Boston-Celtics game

```
library(ggplot2)
ggplot(nba2009_2018, aes(x=home.PTS))+geom_histogram()+xlim(c(50,155))+
  labs(x="Home Points", title="Histogram for points per game by home team")
```



Predicting with distributions

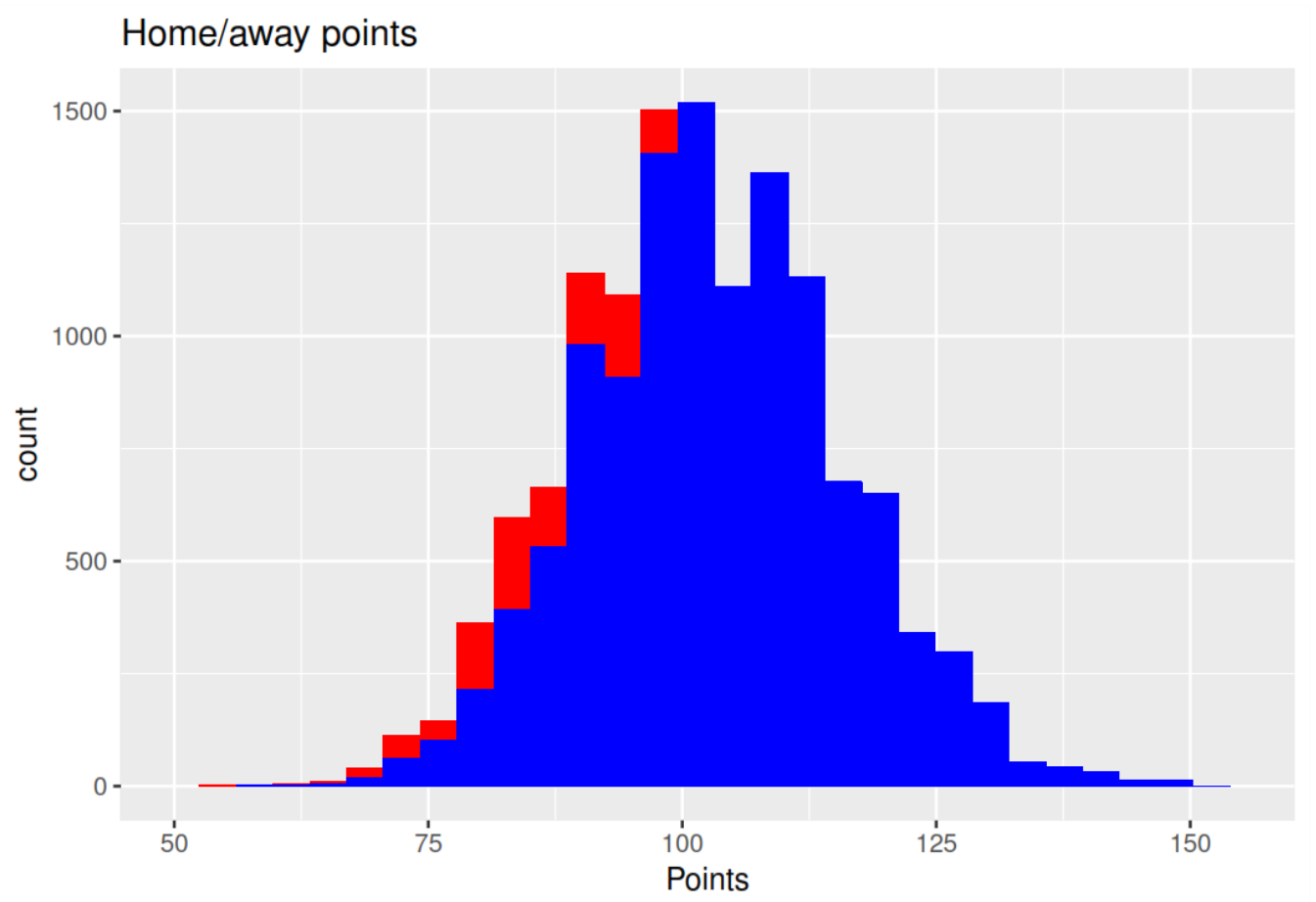
```
ggplot(nba2009_2018, aes(x=away.PTS))+geom_histogram()+xlim(c(50,150))+
  labs(x="Away Points", title="Histogram for points per game by away team")
```



Predicting with distributions

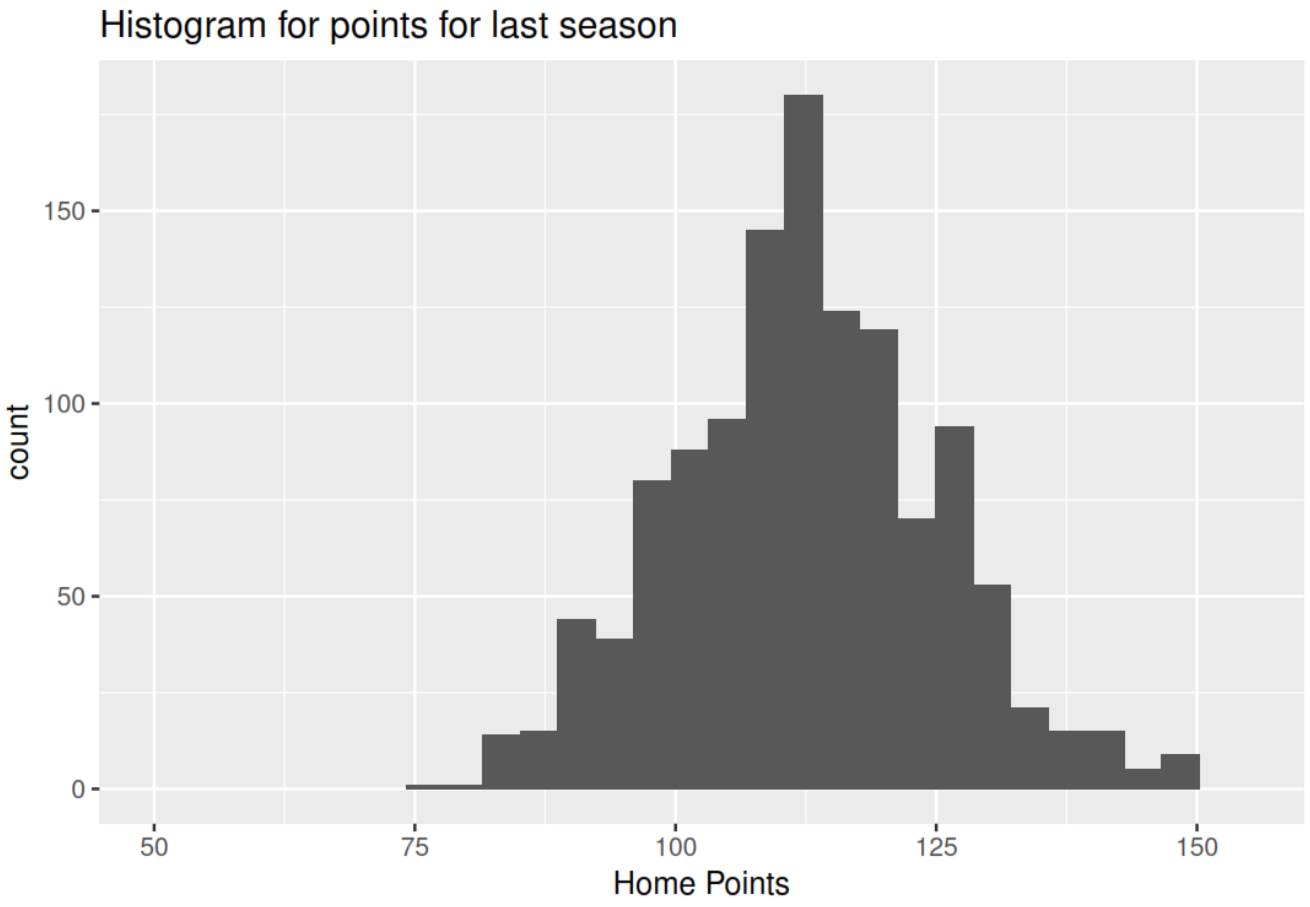
Histogram of home and away points together

```
ggplot(nba2009_2018)+
  geom_histogram(aes(x=away.PTS), fill='red')+
  geom_histogram(aes(x=home.PTS),fill='blue')+
  xlim(c(50,155))+
  labs(x="Points", title="Home/away points")
```



Predicting with distributions

```
library(dplyr)
nba2009_2018 %>%
  filter(SEASON_ID=='2018') %>%
ggplot(aes(x=home.PTS))+geom_histogram()+xlim(c(50,155))+
  labs(x="Home Points", title="Histogram for points for last season")
```

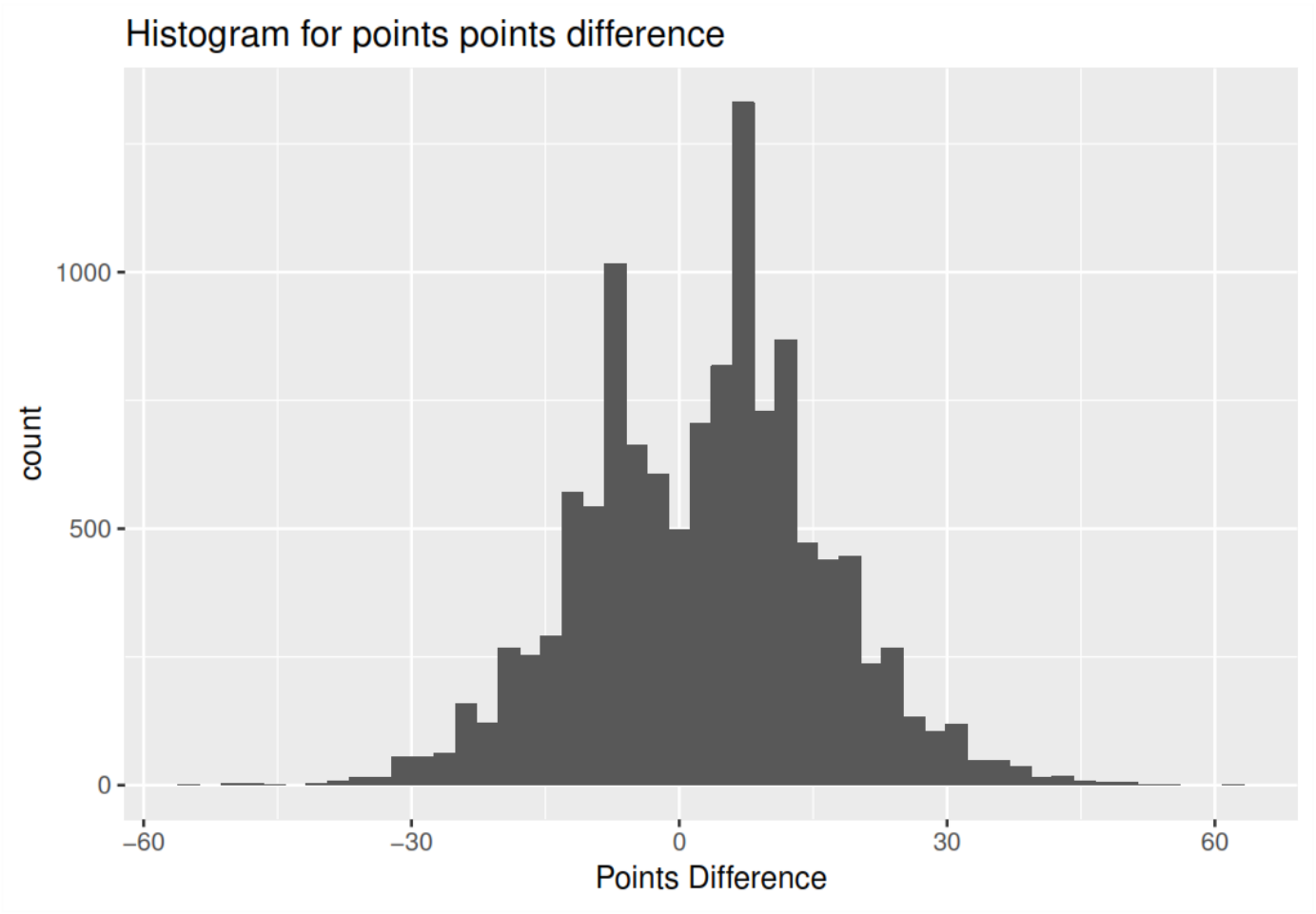


Home points for last season only

Predicting with distributions

What about Points difference ?

```
nba2009_2018$Diff <- nba2009_2018$home.PTS-nba2009_2018$away.PTS
ggplot(nba2009_2018, aes(x=Diff))+geom_histogram(bins=50)+
  labs(x="Points Difference", title="Histogram for points points difference")
```



Predicting with distributions

Test hypothesis of Normality:

$$H_0: sample\ came\ from\ normal\ distribution$$
$$H_1: sample\ is\ not\ from\ normal\ distribution$$

Find the p-value of the test: if is more then $\alpha(0.05)$ then it has a normal distribution, failed to reject H_0

Predicting with distributions

Home points: violation for only 2 teams

```
nba2009_2018 %>%
  filter(SEASON_ID=='2018') %>%
  group_by(home.TEAM_ABBREVIATION) %>%
  summarise(p_value = shapiro.test(home.PTS)$p.value)%>%
  arrange(p_value)
```

```
## # A tibble: 30 x 2
##    home.TEAM_ABBREVIATION p_value
##    <chr>                    <dbl>
##  1 SAS                     0.0330
##  2 TOR                     0.0416
##  3 CLE                     0.0558
##  4 CHA                     0.0582
##  5 IND                     0.0730
##  6 PHI                     0.0759
##  7 MEM                     0.120
##  8 ORL                     0.185
##  9 HOU                     0.189
## 10 MIL                     0.211
## # ... with 20 more rows
```

Predicting with distributions

Away points: Normal distribution with $\alpha = 0.05$

```
nba2009_2018 %>%
  filter(SEASON_ID=='2018') %>%
  group_by(away.TEAM_ABBREVIATION) %>%
  summarise(p_value = shapiro.test(away.PTS)$p.value)%>%
  arrange(p_value)
```

```
## # A tibble: 30 x 2
##    away.TEAM_ABBREVIATION   p_value
##    <chr>                      <dbl>
##  1 CHI                    0.0000237
##  2 MIA                    0.0144
##  3 BKN                    0.0317
##  4 MEM                    0.0406
##  5 TOR                    0.0447
##  6 BOS                    0.187
##  7 PHX                    0.197
##  8 DEN                    0.252
##  9 ORL                    0.277
## 10 MIL                    0.327
## # ... with 20 more rows
```

Predicting with distributions

Suppose we have two variables with normal distribution,

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Then the difference between the home and away teams scores will also have normal distribution with the following parameters (assuming this two are independent)

$$\mu_D = \mu_H - \mu_A$$
$$\sigma_D^2 = \sigma_H^2 + \sigma_A^2$$

Thus the difference will have a normal distribution with the following parameters

$$D \sim N\left(\mu_H - \mu_A, \sigma_H^2 + \sigma_A^2\right)$$

Predicting with distributions

Create dataframes with means and standard deviations for each team, for the last season

```
home <- nba2009_2018 %>%
        filter(SEASON_ID=='2018') %>%
        group_by(home.TEAM_ABBREVIATION) %>%
        summarise(mean = mean(home.PTS),
                  var  = var(home.PTS))
```

```
away <- nba2009_2018 %>%
        filter(SEASON_ID=='2018') %>%
        group_by(away.TEAM_ABBREVIATION) %>%
        summarise(mean = mean(away.PTS),
                  var  = var(away.PTS))
```

Predicting with distributions

Dallas Mavericks playing @ Washington Wizards

Wizards are the home team, Mavericks are the away team



@



Predicting with distributions

Calculate the mean and standard deviation for the points difference between these two teams

$$s_D = \sqrt{\sigma_H^2 + \sigma_A^2}$$

```
mean <- home$mean[home$home.TEAM_ABBREVIATION=="WAS"]-
  away$mean[away$away.TEAM_ABBREVIATION=="DAL"]

sd <- sqrt(home$var[home$home.TEAM_ABBREVIATION=="WAS"]+
  away$var[away$away.TEAM_ABBREVIATION=="DAL"])

mean
```

```
## [1] 9
```

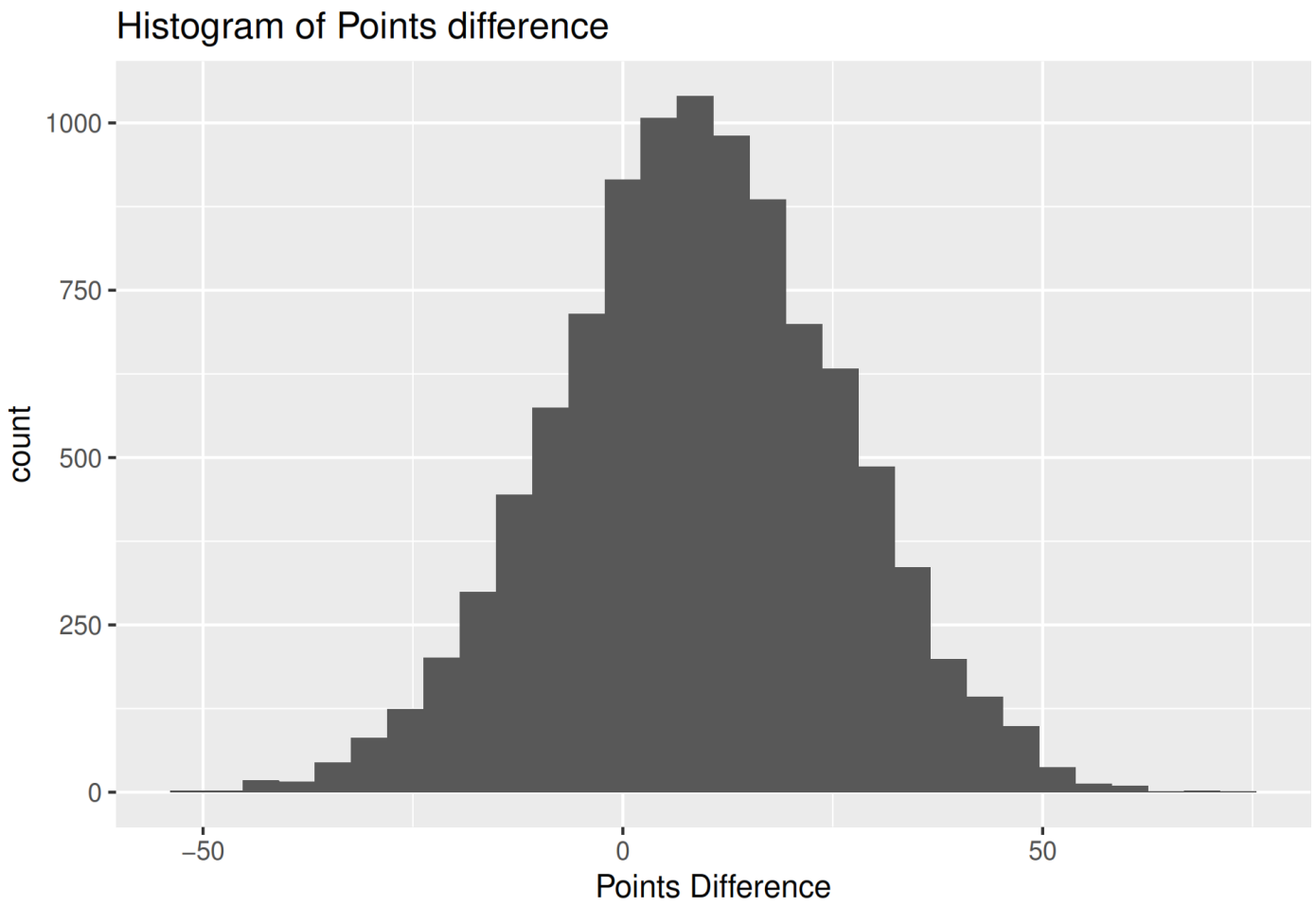
```
sd
```

```
## [1] 16.7122
```

Predicting with distributions

Simulate 10,000 games and plot the histogram.
Can you see the home team advantage ?

```
set.seed(1)
diff <- rnorm(10000, mean, sd)
ggplot()+geom_histogram(aes(x=diff))+
  labs(x="Points Difference", title="Histogram of Points difference")
```



Predicting with distributions

Now lets calculate the probabilities of home and away teams to win

- Away team to win $P(x < 0)$
- Home team to win $P(x > 0)$

Probabiltiy of away team to win

```
pnorm(0,mean,sd, lower.tail = T)
```

```
## [1] 0.295106
```

Probabiltiy of home team to win

```
pnorm(0,mean,sd, lower.tail = F)
```

```
## [1] 0.704894
```

Predicting with distributions

Probability of home team to win with more then 15 points difference

Home team will win with spread of 15 points

```
pnorm(15,mean,sd, lower.tail = F)
```

```
## [1] 0.3597904
```

Predicting with distributions

You can define a function to calculate probabilities for different teams

```
home_away <- list(home,away)
probs <- function(data, home, away, spread=0){
  home_df <- data[[1]]
  away_df <- data[[2]]
  mean_home <- home_df$mean[home_df$home.TEAM_ABBREVIATION==home]
  mean_away <- away_df$mean[away_df$away.TEAM_ABBREVIATION==away]
  mean_diff <- mean_home-mean_away
  var_home <- home_df$var[home_df$home.TEAM_ABBREVIATION==home]
  var_away <- away_df$var[home_df$home.TEAM_ABBREVIATION==away]
  sd_diff <- sqrt(var_home+var_away)
  df <- data.frame(pnorm(spread, mean_diff, sd_diff, lower.tail = F),
                   pnorm(spread, mean_diff, sd_diff))
  colnames(df) <- c(home, away)
  return(df)
}
```

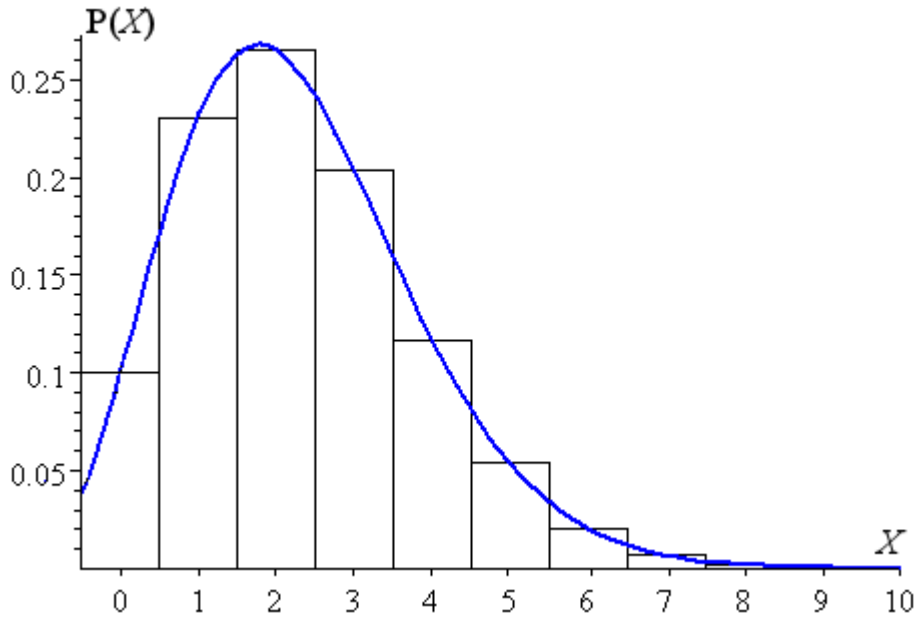
Predicting with distributions

```
probs(home_away, home="WAS", away="DAL", 0)
```

```
##         WAS      DAL
## 1 0.704894 0.295106
```

Predicting football games results

Lets recall what is Poisson distribution about



$$\lambda = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$f(x;\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$$

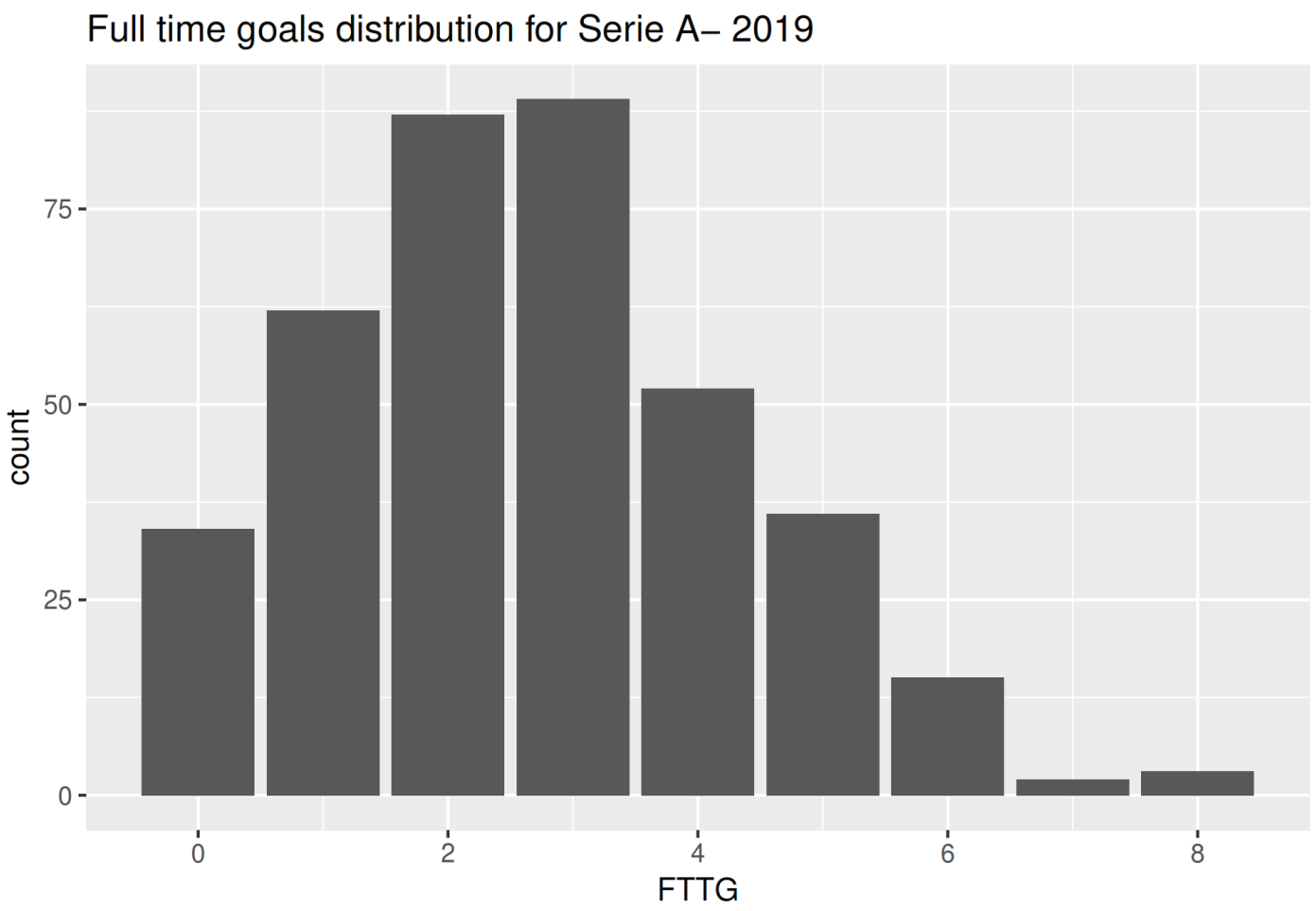
Dataset

```
library(SportsAnalytics270)
library(ggplot2)
data(f_data_sm)
head(f_data_sm)
```

```
##   SEASON COUNTRY         LEAGUE       DATE    HOMETEAM       AWAYTEAM FTSC
## 1   1994 England Premier League 1993-08-14     Arsenal       Coventry  0-3
## 2   1994 England Premier League 1993-08-14 Aston Villa            QPR  4-1
## 3   1994 England Premier League 1993-08-14     Chelsea      Blackburn  1-2
## 4   1994 England Premier League 1993-08-14   Liverpool Sheffield Weds  2-0
## 5   1994 England Premier League 1993-08-14    Man City          Leeds  1-1
## 6   1994 England Premier League 1993-08-14   Newcastle      Tottenham  0-1
##   FTR FTHG FTAG FTTG  H  D  A
## 1   A    0    3    3 NA NA NA
## 2   H    4    1    5 NA NA NA
## 3   A    1    2    3 NA NA NA
## 4   H    2    0    2 NA NA NA
## 5   D    1    1    2 NA NA NA
## 6   A    0    1    1 NA NA NA
```

Predicting with distributions

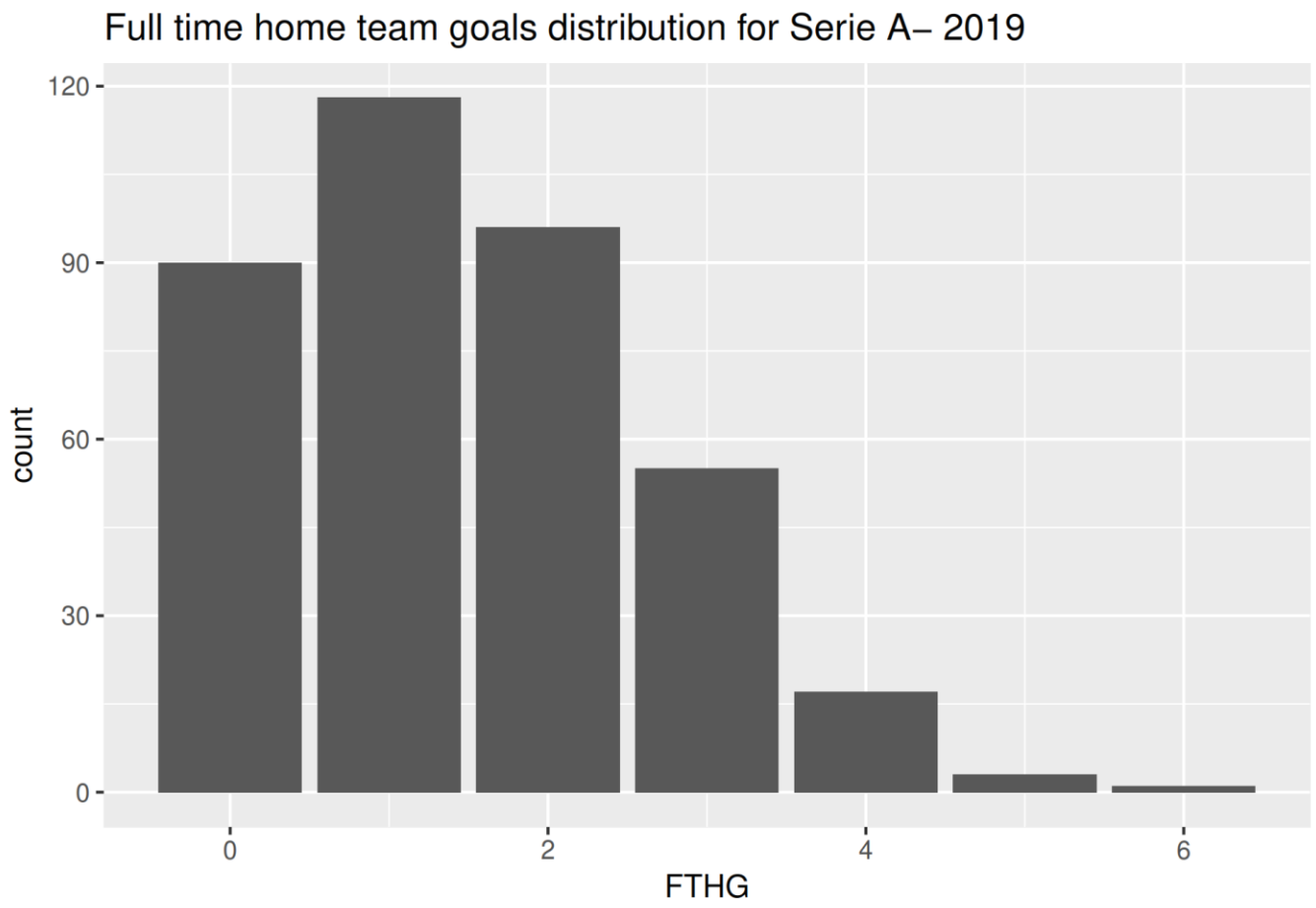
```
f_data_sm %>%
  filter(COUNTRY=="Italy", SEASON=="2019") %>%
ggplot(aes(x=FTTG))+geom_bar()+
ggtitle("Full time goals distribution for Serie A- 2019")
```



Full time total goals distribution for Serie A, Season 2017-2018

Predicting with distributions

```
f_data_sm %>%
  filter(COUNTRY=="Italy", SEASON=="2019") %>%
ggplot(aes(x=FTHG))+geom_bar()+
ggtitle("Full time home team goals distribution for Serie A- 2019")
```



Distribution of home team full time goals

Predicting with distributions

Does the total number of goals scored follows Poisson distribution?

First fine the λ (average number of goals per game)

```
f_data_sm %>%
  filter(COUNTRY=="Italy", SEASON=='2019') %>%
  summarise(mean(FTTG))
```

```
##   mean(FTTG)
## 1   2.681579
```

Predicting with distributions

We can test the hypothesis, whether the goal's distribution is Poisson: Our hypothesis is the following:

$$H_0: The\ variable\ has\ Poisson\ distribution$$
$$H_1: The\ variable\ does\ not\ have\ Poisson\ distribution$$

We will test the hypothesis with Chi-square test: thus comparing empirical probability distribution (from the data) and hypothethical Poisson distribution

Goal frequencies

```
seriea <- f_data_sm %>%
          filter(COUNTRY=="Italy", SEASON=="2019") %>%
          group_by(FTHG) %>%
          summarise(count = n()) %>%
          as.data.frame()
seriea
```

```
##   FTHG count
## 1    0    90
## 2    1   118
## 3    2    96
## 4    3    55
## 5    4    17
## 6    5     3
## 7    6     1
```

Predicting with distributions

We will use chi-square test

First create a vector with observed relative frequencies

```
rel_fr<-seriea$count/sum(seriea$count)
rel_fr
```

```
## [1] 0.236842105 0.310526316 0.252631579 0.144736842 0.044736842 0.007894737
## [7] 0.002631579
```

Then create expected relative frequencies (probabilities), under the assumption that the variable follows Poisson distribution with $\lambda = 1.4842$

```
f_data_sm %>%
   filter(COUNTRY=="Italy", SEASON=="2019") %>%
   summarize(mean(FTHG))
```

```
##   mean(FTHG)
## 1   1.484211
```

```
probs<-dpois(c(0:6), lambda=1.484211)
probs
```

```
## [1] 0.226681122 0.336442614 0.249675914 0.123523913 0.045833888 0.013605432
## [7] 0.003365555
```

Predicting with distributions

Is there a significant difference between these two ?

```
chisq.test(rel_fr, p=probs, rescale.p = T)
```

```
##
##  Chi-squared test for given probabilities
##
## data:  rel_fr
## X-squared = 0.0087047, df = 6, p-value = 1
```

Large p-value talks in favor for NULL hypothesis, thus we have a Poisson distribution

Lazio (Home team) vs. Fiorentina (Away team)



vs



Predicting with distributions

- We assume that probabilities for Lazio and Fiorentina scoring are independent from each other.
- Thus: $P(L = 0 \cap F = 0) = P(L = 0) * P(F = 0) \ldots\ldots$
- If you want to find the probability of the draw, then
$$P(L = 0 \cap F = 0 \cup L = 1 \cap F = 1 \cup L = 2 \cap F = 2 \ldots\ldots)$$
- As $\{L = 0 \cap F = 0, L = 1 \cap F = 1, L = 2 \cap F = 2\}$ events are mutually exclusive, the probability of the draw can be calculated as.

$$P(L = 0 \cap F = 0) + P(L = 1 \cap F = 1) + P(L = 2 \cap F = 2) \ldots\ldots =$$
$$P(L = 0) * P(F = 0) + P(L = 1) * P(F = 1) \ldots.$$

Predicting with distributions

Means (λ parameters) for home and away teams

```
home_means <- f_data_sm %>%
                filter(COUNTRY=="Italy", SEASON=="2019") %>%
                group_by(HOMETEAM) %>%
                summarise(mean = mean(FTHG))

away_means <- f_data_sm %>%
                filter(COUNTRY=="Italy", SEASON=="2019") %>%
                group_by(AWAYTEAM) %>%
                summarise(mean = mean(FTAG))
```

Predicting with distributions

For Lazio as home team and Fiorentina as away team

```
home_means$mean[home_means$HOMETEAM=="Lazio"]
```

```
## [1] 1.894737
```

```
away_means$mean[away_means$AWAYTEAM=="Fiorentina"]
```

```
## [1] 1.157895
```

Predicting with distributions

Generate vector of probabilities for number of goals scored for both teams.

- range from 0 to 9

```
options(scipen=999)
goal_probs_lazio<-dpois(c(0:9), lambda=1.894737)
goal_probs_fior<-dpois(c(0:9), lambda=1.157895)
goal_probs_lazio
```

```
##  [1] 0.1503578740 0.2848886270 0.2698945113 0.1704597055 0.0807440778
##  [6] 0.0305977583 0.0096624508 0.0026154004 0.0006194370 0.0001304078
```

```
goal_probs_fior
```

```
##  [1] 0.31414676431 0.36374896766 0.21059155546 0.08128096970 0.02352870710
##  [6] 0.00544875446 0.00105151426 0.00017393473 0.00002517477 0.00000323886
```

Predicting with distributions

Multiplying two vectors of probabilities we will get the matrix of probabilities for a given result

```
a<-goal_probs_lazio %*% t(goal_probs_fior)
print(round(a, digits=4))
```

```
##          [,1]   [,2]   [,3]   [,4]   [,5]   [,6]   [,7] [,8] [,9] [,10]
##  [1,] 0.0472 0.0547 0.0317 0.0122 0.0035 0.0008 0.0002    0    0     0
##  [2,] 0.0895 0.1036 0.0600 0.0232 0.0067 0.0016 0.0003    0    0     0
##  [3,] 0.0848 0.0982 0.0568 0.0219 0.0064 0.0015 0.0003    0    0     0
##  [4,] 0.0535 0.0620 0.0359 0.0139 0.0040 0.0009 0.0002    0    0     0
##  [5,] 0.0254 0.0294 0.0170 0.0066 0.0019 0.0004 0.0001    0    0     0
##  [6,] 0.0096 0.0111 0.0064 0.0025 0.0007 0.0002 0.0000    0    0     0
##  [7,] 0.0030 0.0035 0.0020 0.0008 0.0002 0.0001 0.0000    0    0     0
##  [8,] 0.0008 0.0010 0.0006 0.0002 0.0001 0.0000 0.0000    0    0     0
##  [9,] 0.0002 0.0002 0.0001 0.0001 0.0000 0.0000 0.0000    0    0     0
## [10,] 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000    0    0     0
```

Predicting with distributions

The probability that the game will end 0-0

```
a<-goal_probs_lazio %*% t(goal_probs_fior)
print(round(a, digits=4))
```

```
##          [,1]   [,2]   [,3]   [,4]   [,5]   [,6]   [,7] [,8] [,9] [,10]
##  [1,] 0.0472 0.0547 0.0317 0.0122 0.0035 0.0008 0.0002    0    0     0
##  [2,] 0.0895 0.1036 0.0600 0.0232 0.0067 0.0016 0.0003    0    0     0
##  [3,] 0.0848 0.0982 0.0568 0.0219 0.0064 0.0015 0.0003    0    0     0
##  [4,] 0.0535 0.0620 0.0359 0.0139 0.0040 0.0009 0.0002    0    0     0
##  [5,] 0.0254 0.0294 0.0170 0.0066 0.0019 0.0004 0.0001    0    0     0
##  [6,] 0.0096 0.0111 0.0064 0.0025 0.0007 0.0002 0.0000    0    0     0
##  [7,] 0.0030 0.0035 0.0020 0.0008 0.0002 0.0001 0.0000    0    0     0
##  [8,] 0.0008 0.0010 0.0006 0.0002 0.0001 0.0000 0.0000    0    0     0
##  [9,] 0.0002 0.0002 0.0001 0.0001 0.0000 0.0000 0.0000    0    0     0
## [10,] 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000    0    0     0
```

Predicting with distributions

3-1 Lazio winning (0.711)

```
a<-goal_probs_lazio %*% t(goal_probs_fior)
print(round(a, digits=4))
```

```
##          [,1]   [,2]   [,3]   [,4]   [,5]   [,6]   [,7] [,8] [,9] [,10]
##  [1,] 0.0472 0.0547 0.0317 0.0122 0.0035 0.0008 0.0002    0    0     0
##  [2,] 0.0895 0.1036 0.0600 0.0232 0.0067 0.0016 0.0003    0    0     0
##  [3,] 0.0848 0.0982 0.0568 0.0219 0.0064 0.0015 0.0003    0    0     0
##  [4,] 0.0535 0.0620 0.0359 0.0139 0.0040 0.0009 0.0002    0    0     0
##  [5,] 0.0254 0.0294 0.0170 0.0066 0.0019 0.0004 0.0001    0    0     0
##  [6,] 0.0096 0.0111 0.0064 0.0025 0.0007 0.0002 0.0000    0    0     0
##  [7,] 0.0030 0.0035 0.0020 0.0008 0.0002 0.0001 0.0000    0    0     0
##  [8,] 0.0008 0.0010 0.0006 0.0002 0.0001 0.0000 0.0000    0    0     0
##  [9,] 0.0002 0.0002 0.0001 0.0001 0.0000 0.0000 0.0000    0    0     0
## [10,] 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000    0    0     0
```

Predicting with distributions

Probability of draw: sum all elements on the diagonal

```
a<-goal_probs_lazio %*% t(goal_probs_fior)
print(round(a, digits=4))
```

```
##          [,1]   [,2]   [,3]   [,4]   [,5]   [,6]   [,7] [,8] [,9] [,10]
##  [1,] 0.0472 0.0547 0.0317 0.0122 0.0035 0.0008 0.0002    0    0     0
##  [2,] 0.0895 0.1036 0.0600 0.0232 0.0067 0.0016 0.0003    0    0     0
##  [3,] 0.0848 0.0982 0.0568 0.0219 0.0064 0.0015 0.0003    0    0     0
##  [4,] 0.0535 0.0620 0.0359 0.0139 0.0040 0.0009 0.0002    0    0     0
##  [5,] 0.0254 0.0294 0.0170 0.0066 0.0019 0.0004 0.0001    0    0     0
##  [6,] 0.0096 0.0111 0.0064 0.0025 0.0007 0.0002 0.0000    0    0     0
##  [7,] 0.0030 0.0035 0.0020 0.0008 0.0002 0.0001 0.0000    0    0     0
##  [8,] 0.0008 0.0010 0.0006 0.0002 0.0001 0.0000 0.0000    0    0     0
##  [9,] 0.0002 0.0002 0.0001 0.0001 0.0000 0.0000 0.0000    0    0     0
## [10,] 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000    0    0     0
```

Predicting with distributions

Probability of Lazio winning: Sum of all the probabilities in the lower triangle, without diagonal

```
##          [,1]   [,2]   [,3]   [,4]   [,5]   [,6]   [,7] [,8] [,9] [,10]
##  [1,] 0.0472 0.0547 0.0317 0.0122 0.0035 0.0008 0.0002    0    0     0
##  [2,] 0.0895 0.1036 0.0600 0.0232 0.0067 0.0016 0.0003    0    0     0
##  [3,] 0.0848 0.0982 0.0568 0.0219 0.0064 0.0015 0.0003    0    0     0
##  [4,] 0.0535 0.0620 0.0359 0.0139 0.0040 0.0009 0.0002    0    0     0
##  [5,] 0.0254 0.0294 0.0170 0.0066 0.0019 0.0004 0.0001    0    0     0
##  [6,] 0.0096 0.0111 0.0064 0.0025 0.0007 0.0002 0.0000    0    0     0
##  [7,] 0.0030 0.0035 0.0020 0.0008 0.0002 0.0001 0.0000    0    0     0
##  [8,] 0.0008 0.0010 0.0006 0.0002 0.0001 0.0000 0.0000    0    0     0
##  [9,] 0.0002 0.0002 0.0001 0.0001 0.0000 0.0000 0.0000    0    0     0
## [10,] 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000    0    0     0
```

Predicting with distributions

Probability of Fiorentina winning: Sum of all the probabilities in the upper triangle, without diagonal

```
##          [,1]   [,2]   [,3]   [,4]   [,5]   [,6]   [,7] [,8] [,9] [,10]
##  [1,] 0.0472 0.0547 0.0317 0.0122 0.0035 0.0008 0.0002    0    0     0
##  [2,] 0.0895 0.1036 0.0600 0.0232 0.0067 0.0016 0.0003    0    0     0
##  [3,] 0.0848 0.0982 0.0568 0.0219 0.0064 0.0015 0.0003    0    0     0
##  [4,] 0.0535 0.0620 0.0359 0.0139 0.0040 0.0009 0.0002    0    0     0
##  [5,] 0.0254 0.0294 0.0170 0.0066 0.0019 0.0004 0.0001    0    0     0
##  [6,] 0.0096 0.0111 0.0064 0.0025 0.0007 0.0002 0.0000    0    0     0
##  [7,] 0.0030 0.0035 0.0020 0.0008 0.0002 0.0001 0.0000    0    0     0
##  [8,] 0.0008 0.0010 0.0006 0.0002 0.0001 0.0000 0.0000    0    0     0
##  [9,] 0.0002 0.0002 0.0001 0.0001 0.0000 0.0000 0.0000    0    0     0
## [10,] 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000    0    0     0
```

Predicting with distributions

Probability of game ending in draw

```
sum(diag(a))
```

```
## [1] 0.2236322
```

Probability of Lazio winning

```
sum(a[lower.tri(a, diag=F)])
```

```
## [1] 0.5456192
```

Probability of Fiorentina winning

```
sum(a[upper.tri(a, diag=F)])
```

```
## [1] 0.2307185
```

Predicting with distributions

- Suppose X is a Poisson random variable with mean λ_X that describes the number of the goals the home team scores on average.
- Suppose Y is a Poisson random variable with mean λ_Y that describes the number of the goals the away team scores on average.
- $\boldsymbol{Z = X - Y}$ random variable has a Skellam distribution and $\boldsymbol{Z}$ represents the outcome of the game with respect to the Home team. If $\boldsymbol{Z > 0}$, then the home team has won, if $\boldsymbol{Z = 0}$, then it was a draw, and if $\boldsymbol{Z < 0}$, the home team has lost.

Library for skellam distribution

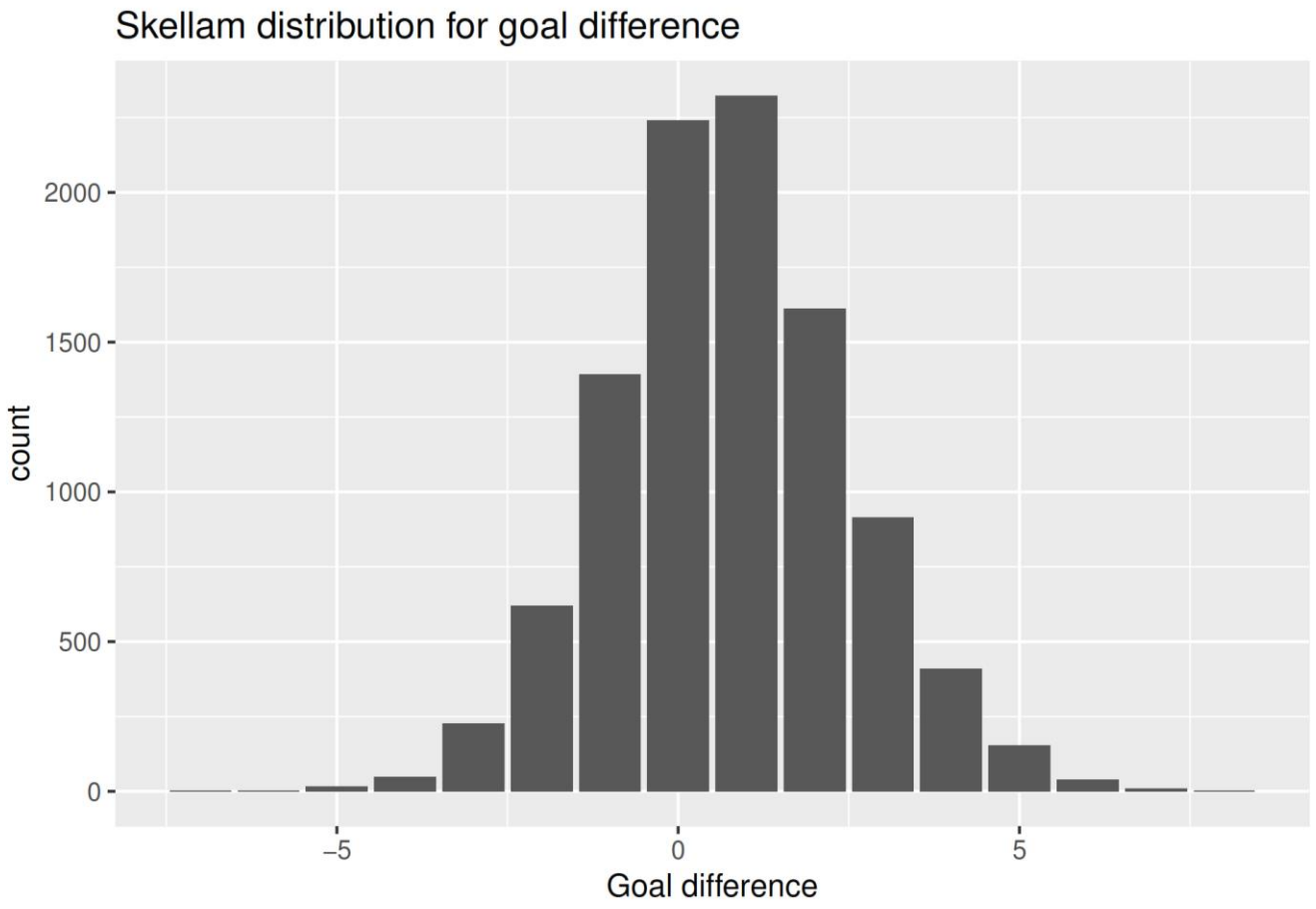
```
install.packages("VGAM")
library(VGAM)
```

Predicting with distributions

Histogram for goal difference

```
set.seed(1)
k <- rskellam(10000,mu1=1.894737, mu2=1.157895)

ggplot()+geom_bar(aes(x=k))+
  labs(x="Goal difference", title="Skellam distribution for goal difference")
```



Predicting with distributions

Calculating probabilities

Probability of home team to win

```
sum(k>0)/10000
```

```
## [1] 0.5458
```

Probability of draw

```
sum(k==0)/10000
```

```
## [1] 0.224
```

Probability of away team to win

```
sum(k<0)/10000
```

```
## [1] 0.2302
```

Predicting with Ratings

Predicting with the rating

- Every team can have some kind of rating, describing its ability to win.
- In football (soccer) the rating can be expressed as the number of points earned, number of goals scored, number of goals conceded, etc.
- One way to go with rating football teams is to calculate goal supremacy for each team.

Goal supremacy = Number of scored goals- Number of conceded goals

Predicting with the rating

- Tottenham playing Leeds United (2002).
- Looking at the goal supremacy (goal difference) for each team for their last 6 games
- Bothe teams will have Goal supremacy rating (we will call it Power Rating) of -3
- What will be your best guess for probability of Tottenham to win? Draw ?

Match	Tottenham	Leeds
1	Arsenal 3:0 Tottenham	Leeds 2:4 Bolton
2	Sunderland 2:0 Tottenham	West Ham 3:4 Leeds
3	Tottenham 0:0 Chelsea	Leeds 0:1 Everton
4	Liverpool 2:1 Tottenham	Middlesbrough 2:2 Leeds
5	Tottenham 3:1 Bolton	Leeds 0:1 Liverpool
6	Blackburn 1:2 Tottenham	Aston Villa 0:0 Leeds

Adapted from: "Fixed Odds Sports Betting" – Joseph Buchdahl

Predicting with the rating

Steps for Power Rating calculation

- Get the Goal Supremacy Rating (Power) for each team n games before the given game.
- Calculate Power Difference for home/away teams – Match rating
- Look at the relative frequencies for home team win, draw and away team win for the given power difference.
- Those will be the probabilities

Predicting with the rating

Kind of table we need

Match rating	No. of home wins	No. of draws	No. of away wins	% of total matches	% of home wins	% of home draws	% of away wins
-11	75	54	91	1.57%	34.1%	24.5%	41.4%
-10	84	94	91	1.92%	31.2%	34.9%	33.8%
-9	123	91	112	2.33%	37.7%	27.9%	34.4%
-8	171	113	124	2.91%	41.9%	27.7%	30.4%
-7	190	121	170	3.44%	39.5%	25.2%	35.3%
-6	242	202	191	4.54%	38.1%	31.8%	30.1%
-5	279	212	197	4.91%	40.6%	30.8%	28.6%
-4	293	219	215	5.19%	40.3%	30.1%	29.6%

Predicting with the rating

- Subset the f_data_sm to get only the games for England
- Select clumns with team names, date, season indicator and final score

```
data(f_data_sm)
england <- f_data_sm %>%
  filter(COUNTRY == "England") %>%
  arrange(DATE) %>%
  select("SEASON","DATE","HOMETEAM",  "AWAYTEAM","FTHG", "FTAG")%>%
  mutate(ID = 1:n())
```

reshape the data into long format

```
england_long <- reshape2::melt(england,
          id.vars = c("ID","HOMETEAM",  "AWAYTEAM",  "SEASON", "DATE"),
          measure.vars = c("FTHG", "FTAG"), variable.name="H_A")

england_long <- england_long[order(england_long$ID),]
rownames(england_long) <- NULL
```

Predicting with the rating

New dataset

```
head(england_long)
```

```
##    ID     HOMETEAM   AWAYTEAM SEASON       DATE  H_A value
## 1   1      Arsenal  Coventry   1994 1993-08-14 FTHG     0
## 2   1      Arsenal  Coventry   1994 1993-08-14 FTAG     3
## 3   2 Aston Villa       QPR   1994 1993-08-14 FTHG     4
## 4   2 Aston Villa       QPR   1994 1993-08-14 FTAG     1
## 5   3      Chelsea Blackburn   1994 1993-08-14 FTHG     1
## 6   3      Chelsea Blackburn   1994 1993-08-14 FTAG     2
```

Predicting with the rating

Make changes in H_A variable

```
england_long$TEAM1 <- ifelse(england_long$H_A =='FTAG',
                             england_long$AWAYTEAM, england_long$HOMETEAM)

england_long$TEAM2 <- ifelse(england_long$H_A =='FTAG',
                             england_long$HOMETEAM, england_long$AWAYTEAM)

england_long <- england_long[,c("ID","SEASON","DATE", "TEAM1","TEAM2", "H_A","value")]
```

Predicting with the rating

```
head(england_long)
```

```
##    ID SEASON       DATE       TEAM1       TEAM2  H_A value
## 1   1   1994 1993-08-14     Arsenal    Coventry FTHG     0
## 2   1   1994 1993-08-14    Coventry     Arsenal FTAG     3
## 3   2   1994 1993-08-14 Aston Villa         QPR FTHG     4
## 4   2   1994 1993-08-14         QPR Aston Villa FTAG     1
## 5   3   1994 1993-08-14     Chelsea   Blackburn FTHG     1
## 6   3   1994 1993-08-14   Blackburn     Chelsea FTAG     2
```

Predicting with the rating

- Calculate the scored and conceded goals for the last 6 games per season for each team
- rollapply from zoo and lag from dplyr are used here
- At the end change the class of the object back to dataframe (from tibble that is default for dplyr)

```
library(dplyr)
library(zoo)
england_long1 <- england_long %>%
group_by(SEASON, TEAM1) %>%
mutate(CUM_SC = dplyr::lag(rollapply(value,6,sum, fill=NA, align='right'),1)) %>%
ungroup() %>%
group_by(SEASON, TEAM2) %>%
mutate(CUM_C = dplyr::lag(rollapply(value,6,sum, fill=NA, align='right'),1)) %>%
filter(CUM_SC != "NA")
england_long1 <- as.data.frame(england_long1)
```

Predicting with the rating

Important:
Round 7 game Arsenal vs Ipswich.

- In last 6 games Arsenal scored 7 goals and conceded 5
- In last 6 games Ipswich scored 7 and conceded 3

```
head(england_long1)
```

```
##    ID SEASON       DATE       TEAM1       TEAM2  H_A value CUM_SC CUM_C
## 1 67   1994 1993-09-11     Arsenal     Ipswich FTHG     4      7     3
## 2 67   1994 1993-09-11     Ipswich     Arsenal FTAG     0      7     5
## 3 68   1994 1993-09-11 Aston Villa    Coventry FTHG     0      9     6
## 4 68   1994 1993-09-11    Coventry Aston Villa FTAG     0     11     5
## 5 69   1994 1993-09-11     Chelsea  Man United FTHG     1      6     3
## 6 69   1994 1993-09-11  Man United     Chelsea FTAG     0     14     6
```

Predicting with the rating

get back to ***wide*** format

```
england_wide <- reshape(england_long1, direction="wide",
                        timevar = "H_A", idvar = "ID",
                        v.names=c("value", "CUM_SC", "CUM_C"))
```

```
head(england_wide)
```

```
##    ID SEASON       DATE       TEAM1      TEAM2 value.FTHG CUM_SC.FTHG
## 1  67   1994 1993-09-11     Arsenal    Ipswich          4           7
## 3  68   1994 1993-09-11 Aston Villa   Coventry          0           9
## 5  69   1994 1993-09-11     Chelsea Man United          1           6
## 7  70   1994 1993-09-11    Man City        QPR          3           5
## 9  71   1994 1993-09-11     Norwich  Wimbledon          0          11
## 11 72   1994 1993-09-11      Oldham    Everton          0           5
##    CUM_C.FTHG value.FTAG CUM_SC.FTAG CUM_C.FTAG
## 1           3          0           7          5
## 3           6          0          11          5
## 5           3          0          14          6
## 7          11          0          10          7
## 9           6          1           8          7
## 11          6          1           7          9
```

Predicting with the rating

Calculate POWERHOME, POWERAWAY and POWERDIFF and RESULT

```
england_wide$POWERHOME <- england_wide$CUM_SC.FTHG-england_wide$CUM_C.FTAG
england_wide$POWERAWAY <- england_wide$CUM_SC.FTAG-england_wide$CUM_C.FTHG
england_wide$POWERDIFF <- england_wide$POWERHOME-england_wide$POWERAWAY

england_wide$RESULT <- ifelse(england_wide$value.FTHG>england_wide$value.FTAG,"HW",
ifelse(england_wide$value.FTHG<england_wide$value.FTAG, "AW", "D"))

england_wide <- england_wide[,c("SEASON","DATE","TEAM1","TEAM2","value.FTHG",
                  "value.FTAG","POWERHOME", "POWERAWAY","POWERDIFF", "RESULT")]
```

Predicting with the rating

Take out remaining NA's

```
england_wide <- england_wide[! is.na(england_wide$POWERDIFF),]
head(england_wide)
```

```
##    SEASON       DATE       TEAM1      TEAM2 value.FTHG value.FTAG
## 1    1994 1993-09-11     Arsenal    Ipswich          4          0
## 3    1994 1993-09-11 Aston Villa   Coventry          0          0
## 5    1994 1993-09-11     Chelsea Man United          1          0
## 7    1994 1993-09-11    Man City        QPR          3          0
## 9    1994 1993-09-11     Norwich  Wimbledon          0          1
## 11   1994 1993-09-11      Oldham    Everton          0          1
##    POWERHOME POWERAWAY POWERDIFF RESULT
## 1          2         4        -2     HW
## 3          4         5        -1      D
## 5          0        11       -11     HW
## 7         -2        -1        -1     HW
## 9          4         2         2     AW
## 11        -4         1        -5     AW
```

Predicting with the rating

Now lets calculate the game result frequencies by Match rating (POWERDIFF)

```
freqs <- dcast(england_wide, POWERDIFF~RESULT)
head(freqs)
```

```
##   POWERDIFF AW D HW
## 1       -30  1 0  0
## 2       -26  2 0  0
## 3       -25  2 0  0
## 4       -24  2 0  0
## 5       -23  1 2  0
## 6       -22  6 0  1
```

Predicting with the rating

Add relative frequencies

```
freqs <- data.frame(freqs, freqs[,2:4]/rowSums(freqs[,2:4]))
head(freqs)
```

```
##   POWERDIFF AW D HW      AW.1      D.1       HW.1
## 1       -30  1 0  0 1.0000000 0.0000000 0.0000000
## 2       -26  2 0  0 1.0000000 0.0000000 0.0000000
## 3       -25  2 0  0 1.0000000 0.0000000 0.0000000
## 4       -24  2 0  0 1.0000000 0.0000000 0.0000000
## 5       -23  1 2  0 0.3333333 0.6666667 0.0000000
## 6       -22  6 0  1 0.8571429 0.0000000 0.1428571
```

Predicting with the rating

Tottenham vs. Leeds United, two teams with equal Goal Supremacy, the POWERDIFF=0

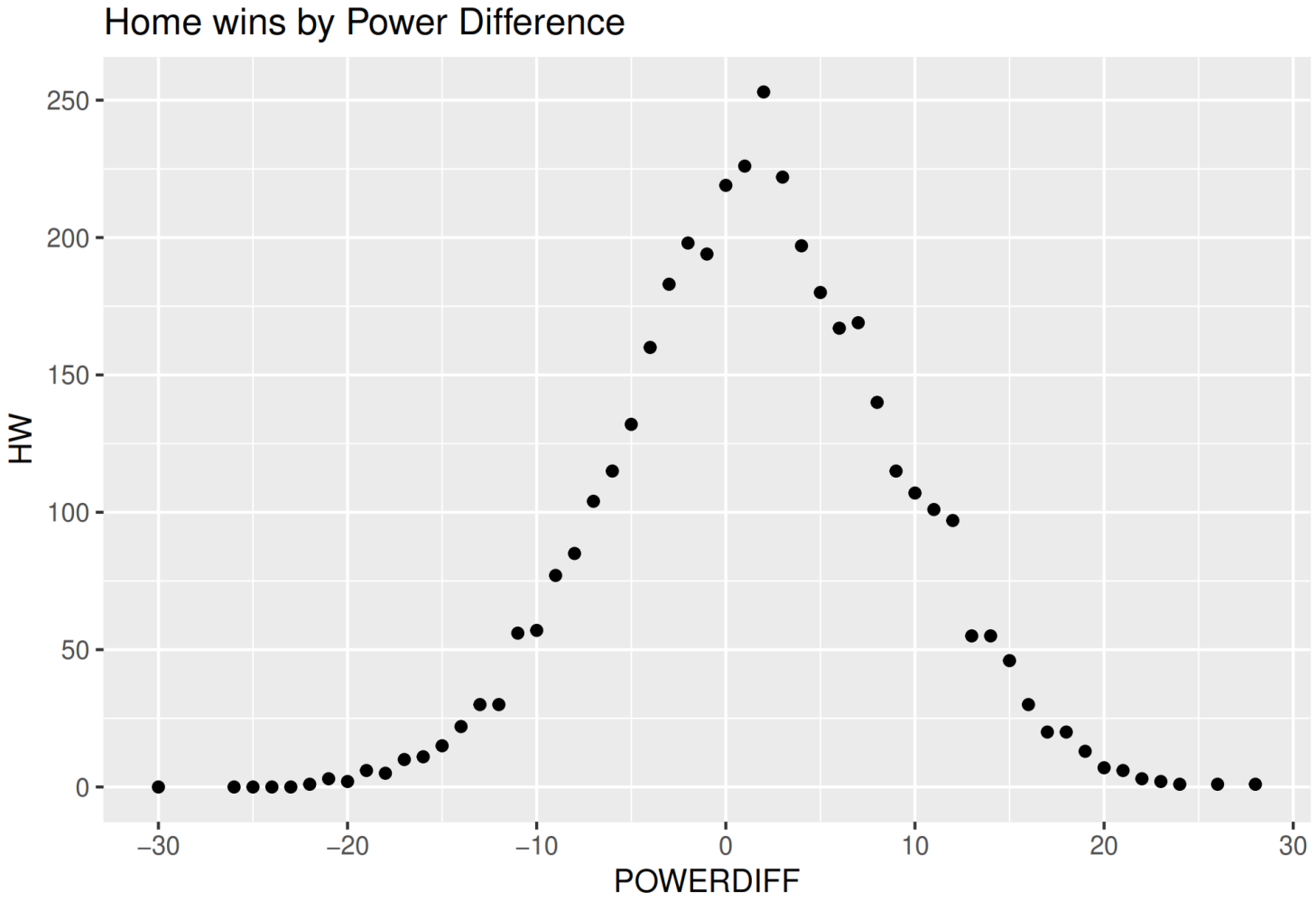
```
freqs[freqs$POWERDIFF==0,]
```

```
##    POWERDIFF  AW   D  HW     AW.1      D.1       HW.1
## 28         0 115 140 219 0.242616 0.2953586 0.4620253
```

Predicting with the rating

Distribution of Home Wins by Power difference

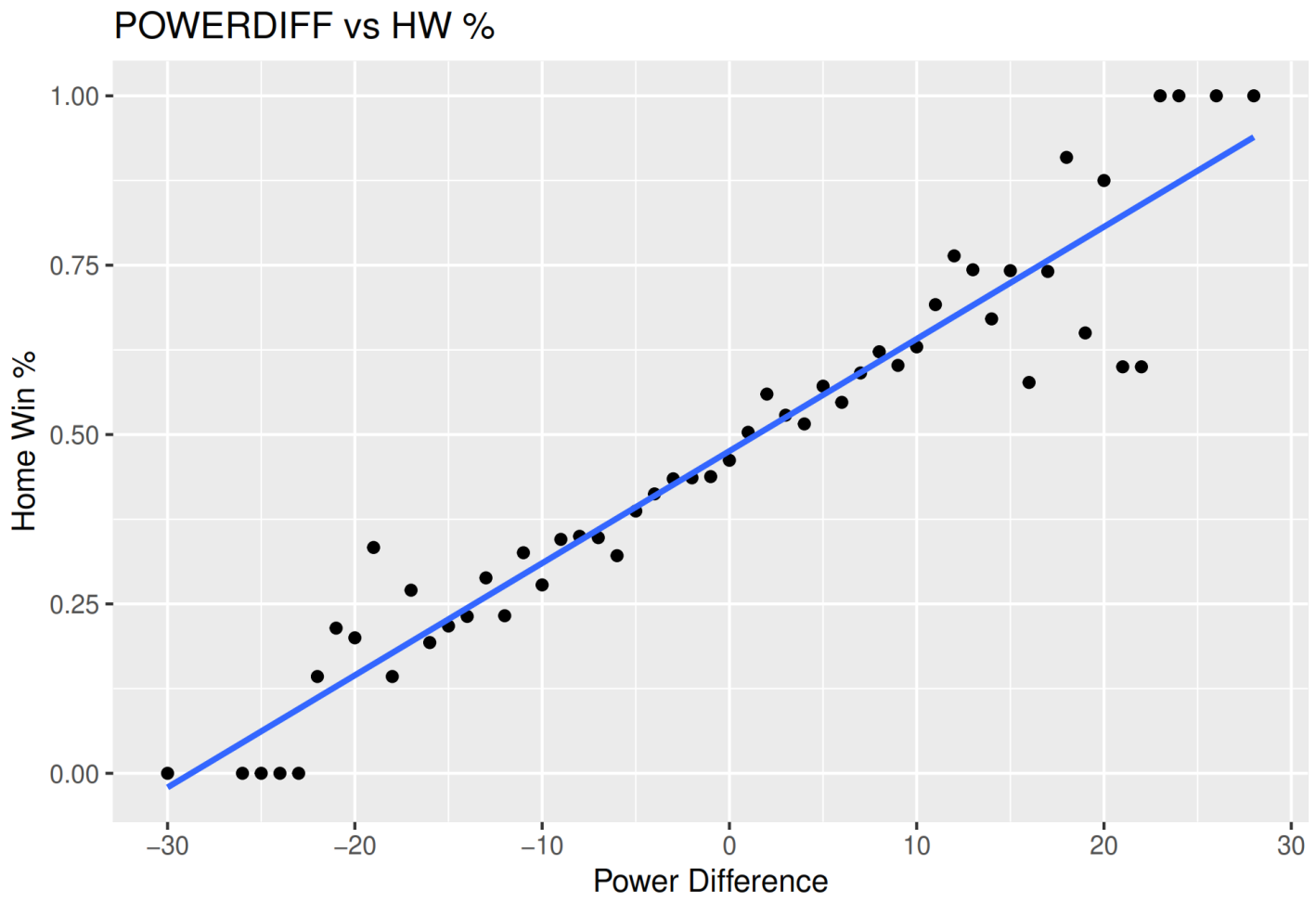
```
ggplot(freqs, aes(x=POWERDIFF, y = HW))+geom_point()+
  ggtitle("Home wins by Power Difference")
```



Predicting with the rating

Look at the relationship between probability of home team to win and power difference, linear with exception of some outliers

```
ggplot(freqs, aes(x=POWERDIFF, y=HW.1))+geom_point()+
  geom_smooth(method="lm", se=F)+
  labs(x="Power Difference", y="Home Win %", title = "POWERDIFF vs HW %")
```



Predicting with the rating

Look at the model

```
modelH <- lm(HW.1~POWERDIFF, data=freqs)
summary(modelH)
```

```
##
## Call:
## lm(formula = HW.1 ~ POWERDIFF, data = freqs)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.239750 -0.027099  0.001157  0.033549  0.172117
##
## Coefficients:
##             Estimate Std. Error t value            Pr(>|t|)
## (Intercept) 0.4756591  0.0107550   44.23 <0.0000000000000002 ***
## POWERDIFF   0.0165496  0.0006812   24.30 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.07899 on 52 degrees of freedom
## Multiple R-squared:  0.919,  Adjusted R-squared:  0.9175
## F-statistic: 590.3 on 1 and 52 DF,  p-value: < 0.00000000000000022
```

Predicting with the rating

Away team probability

```
modelA <- lm(AW.1~POWERDIFF, data=freqs)
summary(modelA)
```

```
##
## Call:
## lm(formula = AW.1 ~ POWERDIFF, data = freqs)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.33810 -0.06875 -0.03910  0.02951  0.41701
##
## Coefficients:
##              Estimate Std. Error t value             Pr(>|t|)
## (Intercept)  0.319560   0.017716   18.04 <0.0000000000000002 ***
## POWERDIFF   -0.015299   0.001122  -13.63 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1301 on 52 degrees of freedom
## Multiple R-squared:  0.7814, Adjusted R-squared:  0.7772
## F-statistic: 185.9 on 1 and 52 DF,  p-value: < 0.00000000000000022
```

Predicting with the rating

Draw: bad model

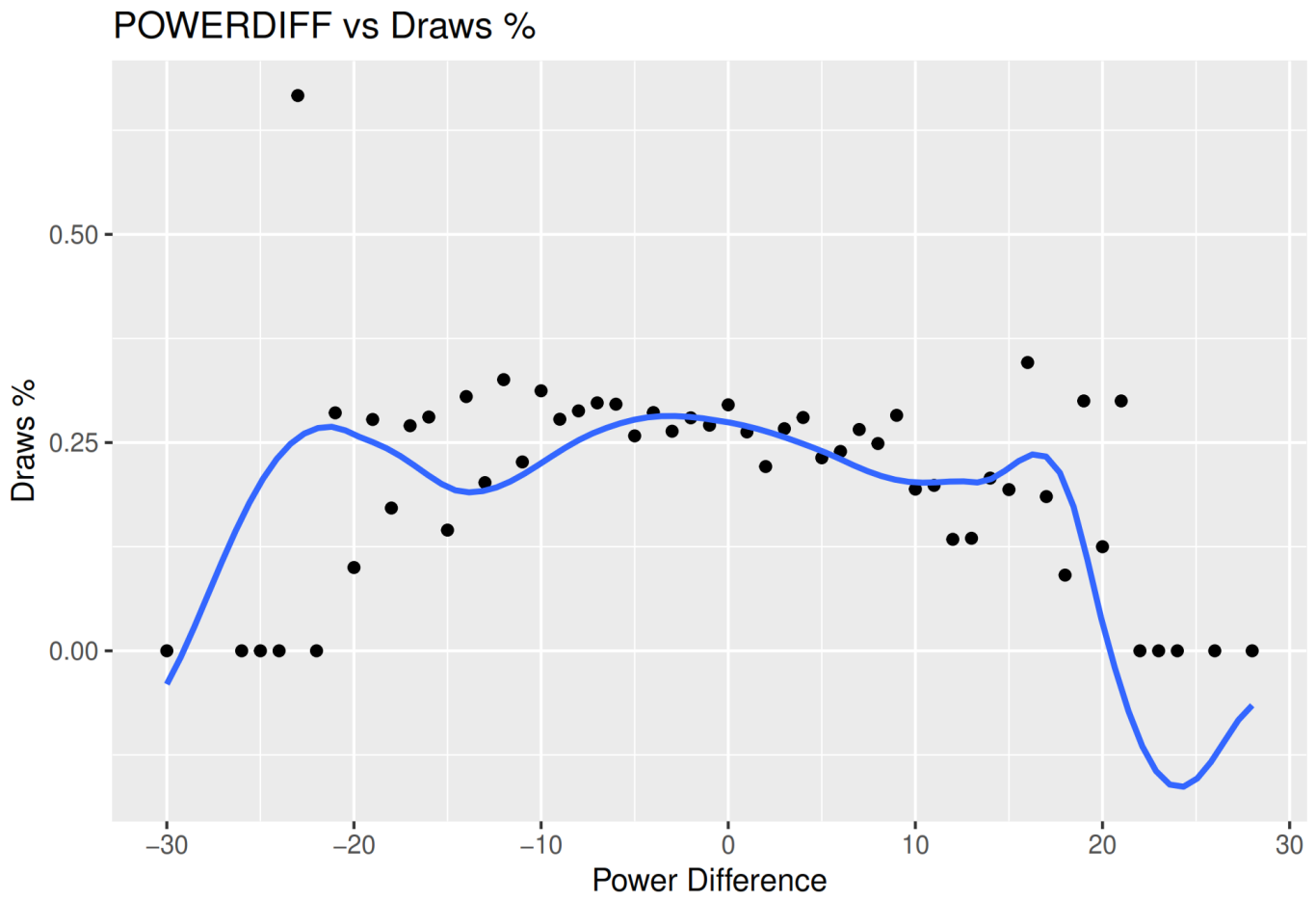
```
modelD <- lm(D.1~POWERDIFF, data=freqs)
summary(modelD)
```

```
##
## Call:
## lm(formula = D.1 ~ POWERDIFF, data = freqs)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.24231 -0.05590  0.04315  0.07307  0.43312
##
## Coefficients:
##             Estimate Std. Error t value             Pr(>|t|)
## (Intercept)  0.204781   0.017316  11.826 0.0000000000000232 ***
## POWERDIFF   -0.001251   0.001097  -1.141                0.259
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1272 on 52 degrees of freedom
## Multiple R-squared:  0.02441,    Adjusted R-squared:  0.005644
## F-statistic: 1.301 on 1 and 52 DF,  p-value: 0.2593
```

Predicting with the rating

relationship with probability of draw is more like polynomial

```
ggplot(freqs, aes(x=POWERDIFF, y=D.1))+geom_point()+
  geom_smooth(formula=y~poly(x,2), se=F)+
  labs(x="Power Difference", y="Draws %", title = "POWERDIFF vs Draws %")
```



Predicting with the rating

Draw: very hard to predict

```
modelD <- lm(D.1~poly(POWERDIFF,2), data=freqs)
summary(modelD)
```

```
##
## Call:
## lm(formula = D.1 ~ poly(POWERDIFF, 2), data = freqs)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.16319 -0.04788 -0.00960  0.01603  0.51653
##
## Coefficients:
##                     Estimate Std. Error t value             Pr(>|t|)
## (Intercept)          0.20541    0.01395  14.727 < 0.0000000000000002 ***
## poly(POWERDIFF, 2)1 -0.14506    0.10250  -1.415                0.163
## poly(POWERDIFF, 2)2 -0.55259    0.10250  -5.391          0.00000181 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1025 on 51 degrees of freedom
## Multiple R-squared:  0.3786, Adjusted R-squared:  0.3542
## F-statistic: 15.54 on 2 and 51 DF,  p-value: 0.000005387
```